

Interval Estimation

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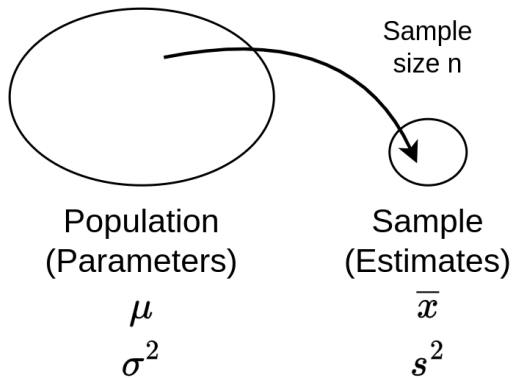
Outline

Sample

Confidence Interval

Lab Report

Sampling & Inference



Sampling

We can use `sample(x, size)` function to sample from population.
size: The number of sample we want

```
# We randomly generate a vector x
x <- rnorm(10000, 3, 0.5)

# We can use sample() to get a sample from x
sample(x, size = 10)
```

Distribution of the Sample Mean

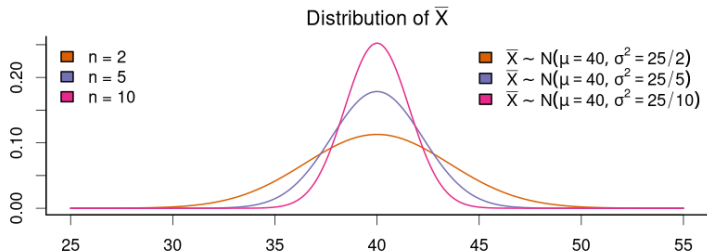
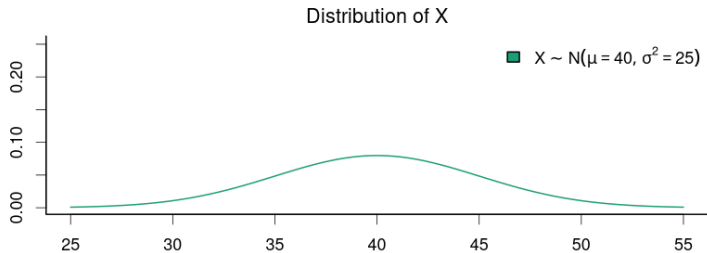
If we draw a sample of size n from a population where each element follows a normal distribution:

$$X \sim N(\mu, \sigma^2)$$

Then the sample mean $\bar{X}$ follows a normal distribution:

$$\bar{X} \sim N\left(\mu_{\bar{X}} = \mu, \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}\right)$$

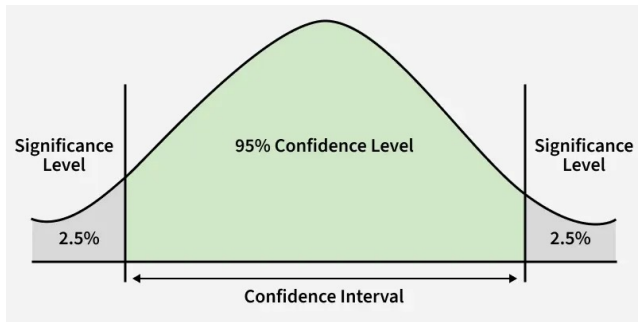
Distribution of mean for various sample sizes



Confidence Interval (σ^2 known)

If population variance σ^2 is known, then we can build CI for the population mean μ

$$\bar{X} - Z_{[1-\alpha/2]} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{[1-\alpha/2]} \frac{\sigma}{\sqrt{n}}$$



Confidence Interval (σ^2 known)

Here we'll use the example mentioned in Steven's lecture.

- ▶ X : Shoulder heights of African elephants
- ▶ Randomly select 29 elephants $\Rightarrow n = 29$
- ▶ Sample mean $\mu_{\bar{x}} = 271$ cm
- ▶ $\sigma = 20.8$ cm

Calculate the 95% CI for the true population mean shoulder height for male elephants.

Confidence Interval (σ^2 known)

First we set the $\mu_{\bar{X}}$, n , and σ that the question gives.

```
mean_value <- 271
n <- 29
standard_deviation <- 20.8
```

Then we calculate the standard error $\frac{\sigma}{\sqrt{n}}$

```
standard_error <- standard_deviation / sqrt(n)
```

Set the significant level α and calculate Z score. Then get the margin error $Z_{1-\alpha} \frac{\sigma}{\sqrt{n}}$

```
alpha <- 0.05
z_score <- qnorm(alpha/2, lower.tail = FALSE)
margin_error <- z_score * standard_error
```

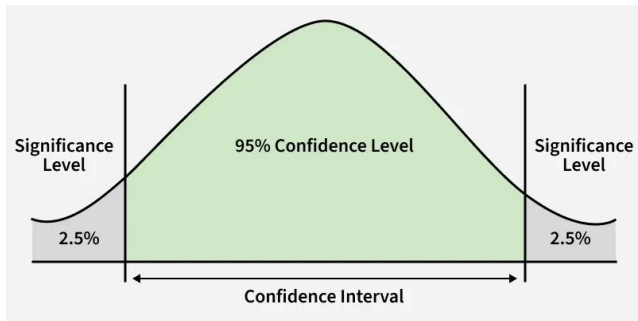
Calculate lower bound $= \bar{X} - Z_{[1-\alpha/2]} \frac{\sigma}{\sqrt{n}}$ and upper bound $= \bar{X} + Z_{[1-\alpha/2]} \frac{\sigma}{\sqrt{n}}$

```
lower_bound <- mean_value - margin_error
upper_bound <- mean_value + margin_error
print(c(lower_bound, upper_bound))
```

Confidence Interval (σ^2 unknown)

When population variance σ^2 is unknown, we can construct $100(1-\alpha)\%$ confidence intervals as before. We shall use s and relevant values from the t distribution instead of using α and the Z distribution

$$\bar{X} - t_{df, \alpha/2} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{df, \alpha/2} \frac{s}{\sqrt{n}}$$



Confidence Interval (σ^2 unknown)

Here we'll also use the example mentioned in Steven's lecture.

- ▶ X : Shoulder heights of African elephants
- ▶ Randomly select 18 elephants $\Rightarrow n = 18$
- ▶ Sample mean $\mu_{\bar{X}} = 229$ cm
- ▶ $s = 15$ cm

Calculate the 95% confidence interval for the true mean shoulder height for this population of female elephants.

Confidence Interval (σ^2 unknown)

First we set the $\mu_{\bar{X}}$, n , and σ that the question gives.

```
mean_value <- 229
n <- 18
standard_deviation <- 15
standard_error <- standard_deviation / sqrt(n)
```

Set the significant level α and calculate t score. Then get the margin error $t_{df,\alpha} \frac{\sigma}{\sqrt{n}}$

```
alpha <- 0.05
degrees_of_freedom <- n - 1
t_score <- qt(alpha, df = degrees_of_freedom, lower.tail =
  FALSE)
margin_error <- t_score * standard_error
```

Calculate lower bound = $\bar{X} - t_{[17,0.05]} \frac{\sigma}{\sqrt{n}}$ and upper bound = $\bar{X} + t_{[17,0.05]} \frac{\sigma}{\sqrt{n}}$

```
lower_bound <- mean_value - margin_error
upper_bound <- mean_value + margin_error
print(c(lower_bound, upper_bound))
```

Central Limit Theorem (CLT)

If X is distributed as a random variable (does NOT need to be normal) with $\mu_x = \mu$ and $\sigma_x^2 = \sigma^2$

then as $n \rightarrow \infty$

$$\bar{X} \sim N(\mu_{\bar{X}} = \mu, \sigma_{\bar{X}}^2 = \sigma^2/n)$$

We can make inferences about the population mean μ regardless of the distribution of X .

As n increase, σ^2/n decreases. $\bar{X}$ is tightly clustered around the population mean μ .

In the following R code, we'll demonstrate how CLT work on certain distribution metioned in the lecture

Lab Report

- ▶ The report file will be in the NTU COOL Assignment section
- ▶ Submit your file in pdf format
- ▶ Deadline: 4/26 (Sun) 23:59
- ▶ Late submission is not accepted